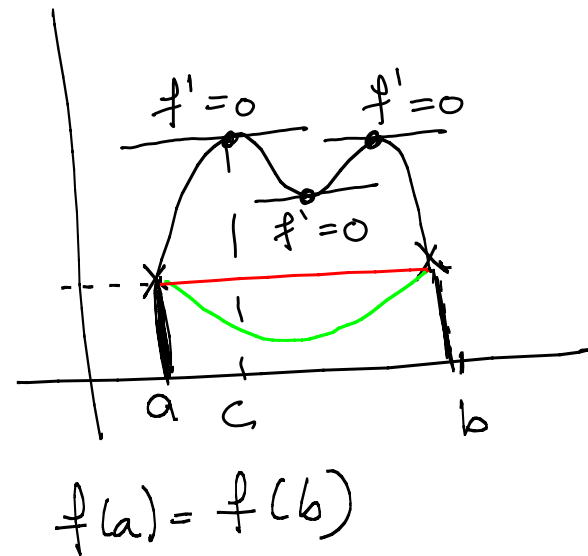


Unit 4.2 Two important theorems

Note Title

2015/04/19

- Theorem of Rolle
- Mean value Theorem.



Rolle's Theorem

I f

1. f is continuous on $[a, b]$
2. f is differentiable on (a, b)
3. $f(a) = f(b)$

then there exists a $c \in (a, b)$ such that $f'(c) = 0$

Example: Find c of Rolle's Theorem for $f(x) = x^3 - x + 1$ on $[0, 1]$, if such a c exists.

- Conditions:
1. f is cont on $[0, 1]$ - polynomial
 2. f is db on $(0, 1)$ - polynomial
 3. $f(0) = 1 = f(1)$

So there exist a $c \in (0, 1)$ such that $f'(c) = 0$

Now

$$f'(x) = 3x^2 - 1$$

$$\Rightarrow f'(c) = 3c^2 - 1 = 0$$

$$\Rightarrow c^2 = \frac{1}{3}$$

$$\Rightarrow c = \frac{1}{\sqrt{3}} \text{ or } c = -\frac{1}{\sqrt{3}}$$

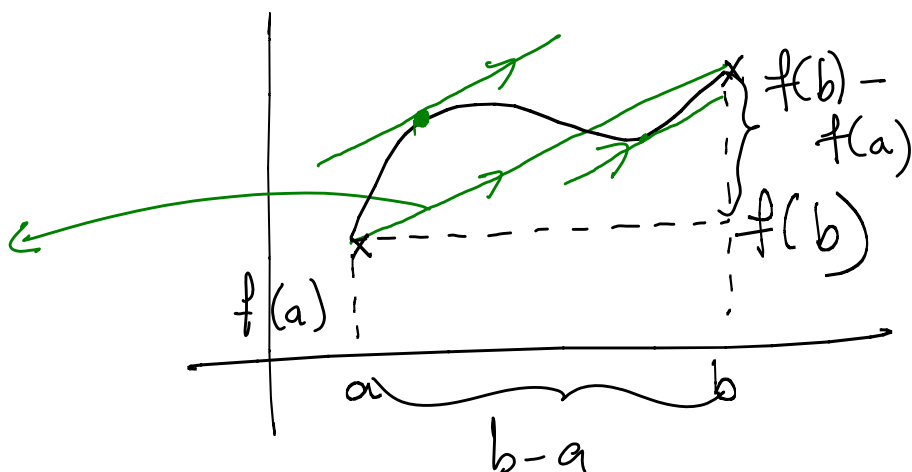
Not in interval

$$\text{So } c = \frac{1}{\sqrt{3}}$$

The Mean value theorem

Gradient of
connecting line
is

$$= \frac{f(b) - f(a)}{b - a}$$



Mean Value Theorem

If

1. f is continuous on $[a, b]$
2. f is differentiable on (a, b)

then there exists a $c \in (a, b)$ such
that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Ex. Find c of the MVT for $f(x) = x^2 + 1$ on $[-1, 2]$.

1. f is cont on $[-1, 2]$ (Polynomial)
2. f is db on $(-1, 2)$ "

So there exists a $c \in (-1, 2)$ such that

$$f'(c) = \frac{f(2) - f(-1)}{2 - (-1)}$$

$$\Rightarrow 2c = \frac{5 - 2}{3} = 1$$

$$\Rightarrow c = \frac{1}{2}$$

